

1. INTRODUCTION

These sources investigate the use of **deep learning** to automate safety monitoring, specifically focusing on **face mask detection**, **social distancing**, and **personal protective equipment (PPE)** compliance. Researchers employ various high-performance architectures such as **MobileNet**, **YOLO**, and **ResNet-50** to process real-time video feeds from public and industrial spaces. By utilizing **computer vision** techniques, these systems can identify individuals, calculate physical distance between them, and verify the presence of essential safety gear like helmets, vests, and gloves. Several papers highlight the integration of **IoT features**, including non-contact temperature measurement and instant smartphone alerts for health violations. The collective findings demonstrate that **transfer learning** and data augmentation significantly enhance the **accuracy** and efficiency of these monitoring tools. Ultimately, these technological advancements aim to reduce labor costs and improve **public health** security during pandemics and in hazardous work environments.

2. FOUNDATIONAL DEEP LEARNING TECHNIQUES

2.1 Deep Residual Learning for Image Recognition (ResNet)

The paper “**Deep Residual Learning for Image Recognition**” (2015), authored by **Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun**, addresses the **degradation problem** in deep learning, where accuracy saturates and then rapidly declines as network depth increases, resulting in higher training errors for deeper models. The key innovation is a **residual learning framework** that employs **identity shortcut connections** to skip one or more layers. Instead of expecting stacked nonlinear layers to directly fit a desired underlying mapping, the methodology explicitly reformulates them to fit a **residual mapping**, recasting the original mapping as . This approach allows the network to be optimized more easily and trained end-to-end via backpropagation without adding extra parameters or computational complexity. The work is highly significant as it enabled the construction of networks up to **152 layers deep**—8x deeper than VGG—which achieved state-of-the-art results and won **1st place in the ILSVRC 2015** classification task and several COCO 2015 competitions, proving that depth is essential for effective feature representation.

2.2 You Only Look Once: Unified Real-Time Object Detection (YOLO)

In the paper “**You Only Look Once: Unified, Real-Time Object Detection**” (2016), authors **Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi** address the limitations of traditional object detection systems that repurpose classifiers (such as R-CNN or sliding window approaches), which are slow and difficult to optimize because they rely on complex, multi-stage pipelines. The key innovation is reframing object detection as a **single regression problem** that maps image pixels directly to bounding box coordinates and class probabilities, allowing for end-to-end optimization.

The methodology works by dividing the input image into an **grid**, where each cell simultaneously predicts multiple bounding boxes, confidence scores for those boxes, and conditional class probabilities using a single convolutional neural network. This work is highly significant to the field because it is **extremely fast**, enabling real-time processing at **45 to 155 frames per second**, and it reduces background errors by reasoning globally about the image context, making it more robust and **generalizable** than prior state-of-the-art methods.

2.3 MobileNets: Efficient Convolutional Neural Networks for Mobile Vision

The paper “**MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications**” (2017), authored by **Andrew G. Howard et al.**, addresses the challenge of deploying deep convolutional neural networks on computationally limited platforms, where traditional architectures are often too slow and large for real-time mobile or embedded use. The key innovation is an efficient architecture built on **depthwise separable convolutions**, which drastically reduce computation and model size by factorizing a standard convolution into two distinct layers: a depthwise convolution for filtering and a pointwise convolution for feature combination. Additionally, the authors introduced two global hyper-parameters—**width multiplier** and **resolution multiplier**—enabling model builders to trade off accuracy for improvements in latency and size based on specific application requirements. This research is highly significant to the field because it provides a scalable framework that allows complex vision tasks to run on-device; for instance, MobileNet achieves ImageNet accuracy similar to VGG16 while being **32 times smaller** and **27 times less compute-intensive**.

3. FACE MASK DETECTION SYSTEMS

3.1 Face Mask and Social Distance Detection Using Deep Learning Models

1. Full Citation Details

- **Authors:** N. Anusha, Saumya Gupta, Y. Nikitha Naidu, M. Ruchitha, and Richal Pandey.
- **Title:** **Face Mask and Social Distance Detection Using Deep Learning Models.**
- **Year:** 2023.
- **Publication Venue:** *Computational Vision and Bio-Inspired Computing* (Advances in Intelligent Systems and Computing, Volume 1439).

2. Specific Problem Identified or Addressed

The researchers addressed the global threat of the **COVID-19 pandemic**, identifying that flattening the curve is difficult unless residents adhere to measures such as wearing masks and social distancing.

They specifically noted that while many studies have focused on mask detection or social distancing separately, there was a **lack of integrated systems** that successfully monitored both violations simultaneously.

3. Introduction/Background Summary

The COVID-19 virus was first identified in Wuhan, China, in December 2019 and spread to 216 countries, causing millions of deaths globally. The virus spreads through **contaminated droplets, aerosols, and small airborne particles**. To halt this spread, governments issued guidelines for wearing facemasks, maintaining social distance, and vaccination. The authors emphasize that adherence to these measures is essential in **high-traffic public areas** like shopping malls, hospitals, and airports.

4. Research Motivation

The motivation was to **encourage public caution and awareness** by developing emerging technologies capable of capturing live or recorded media to detect conformity to health norms. The authors believed that providing automated alerts via **alarms or voice messages** would help people protect themselves and those around them from the disease.

5. Specific Objectives

The research aimed to propose and identify the **most efficient deep learning model** among widely used options to detect face mask and social distancing violations. They sought to implement and compare models to ascertain their performance in identifying people and calculating distances between them using **Euclidean or Manhattan metrics** within a 6-foot threshold.

6. Methodology

- **Deep Learning Architectures/Models Used:** The study implemented and compared **MobileNet, YOLO (v3 and v5), and ResNet-50**.
- **Dataset Details:** The authors utilized multiple datasets, including **Face Mask Lite, Real-World Masked Face (RMFD), MaskedFace-Net, and the Face Mask Detection dataset** from Kaggle. The total dataset size involved approximately **3,800 to 3,853 images**. The classes included “with mask,” “without a mask,” and “mask is worn incorrectly”.
- **Training Approach:** Models were processed in an **Anaconda + Python 3.x environment**. Data was split into various ratios: **90:10, 80:20, 70:30, and 60:40** to evaluate performance. Training included a **batch size of 32 and 10 epochs**. Preprocessing involved scaling images to **224x224 pixels** and converting them into arrays.
- **Evaluation Metrics:** Performance was measured using **precision, recall, accuracy, F1-score, and a confusion matrix**.

7. Main Contributions to Knowledge

- The development of an **integrated approach** that monitors both face mask compliance and social distancing violations in a single system.
- A comprehensive **comparative analysis** of several deep learning models across different training and testing ratios.
- The empirical identification of **MobileNet** as the superior model for this specific application, achieving the highest accuracy of **99%** compared to ResNet-50 and YOLOv5.

8. Limitations

The sources mention several disadvantages and challenges observed during the survey and experimentation:

- Difficulty in recognizing **blurry images** and faces obscured by **hands or scarves**.
- Complications in social distance detection caused by the varying **distance of people from the camera** and a lack of focal length/sensor dimension data.
- The **YOLOv5 model** performed significantly lower than expected, with only **55% accuracy** in their testing scenario.
- Accuracy for even the best models (MobileNet and ResNet-50) slightly decreased by approximately 1% when the training-to-testing ratio was dropped to **60:40**.

3.2 Real-Time Face Mask Detection Using Deep Learning: Enhancing Public Health and Safety

1. Full Citation Details

- **Authors:** Ratnam Dodda, Raghavendra C., Raghavendra Swamy U., Chandu Naik Azmera, Sreenu M, and Satyanarayana Nimmala.
- **Title:** Real-Time Face Mask Detection Using Deep Learning: Enhancing Public Health and Safety.
- **Year:** 2025.
- **Publication Venue:** *ICREGCSD 2025*, published in *E3S Web of Conferences* 616, 02013.

2. Specific Problem Identified or Addressed

The researchers addressed the **logistical challenge of large-scale mask compliance** during global health crises like COVID-19. They identified that **manual surveillance methods are labor-intensive** and highly susceptible to human error. Additionally, they noted that traditional computer vision methods, such as Haar cascades and HOG combined with SVM, were too **rigid to adapt to complex and dynamic environments**.

3. Introduction/Background Summary

The context for this work is the rise of global health crises which necessitated technologies to reinforce health protocols and enhance public safety. Face masks were established as an essential tool for **reducing the transmission of airborne pathogens**. While initial technological solutions used classical machine learning, recent advancements in **Convolutional Neural Networks (CNNs)** have significantly improved accuracy and adaptability in image classification and object detection. The system is designed for deployment in **high-traffic environments** such as airports, hospitals, corporate offices, and schools.

4. Research Motivation

The motivation was to develop a **consistent, non-invasive approach** to enforcing safety measures without relying heavily on personnel for manual monitoring. By automating the detection process, the authors aimed to create **practical tools using AI** that contribute directly to global public health and safety efforts.

5. Specific Objectives

- To develop an **automated system** that accurately identifies individuals with and without masks in **real-time**.
- To create a **scalable and adaptable solution** suitable for diverse settings, including clean rooms, laboratories, and hazardous environments.

- To leverage **cloud-based resources** (Google Colab and Google Drive) to eliminate the need for extensive local hardware while maintaining high performance.

6. Methodology

- **Deep Learning Architecture/Model Used:** The system utilizes a **Convolutional Neural Network (CNN)**. The architecture includes **Conv2D layers** for feature extraction, **MaxPooling2D layers** for down-sampling, and **dense layers** for final classification. It employs **ReLU activation** for hidden layers and **Softmax** for the output layer.
- **Dataset Details:** The system uses a comprehensive dataset of images featuring individuals **wearing and not wearing masks**. Specific data augmentation techniques—**rotation, zoom, and horizontal flipping**—were used to increase dataset diversity and improve generalization.
- **Training Approach:** The model was developed using **TensorFlow and Keras** on the **Google Colab** cloud platform. Images were preprocessed by resizing them to **150x150 pixels** and normalizing pixel values. Training involved a **binary cross-entropy loss function** and the **Adam optimizer**.
- **Evaluation Metrics:** The performance was measured using **training accuracy, validation accuracy, training loss, and validation loss** monitored over 10 epochs.

7. Main Contributions to Knowledge

- The development of an **accessible, cloud-based solution** for real-time mask compliance that simplifies data management and deployment.
- Demonstration of a **robust CNN model** that achieves nearly **100% training accuracy** and over **90% validation accuracy**.
- Providing a **scalable tool** for continuous video monitoring that reduces reliance on human personnel for enforcing health protocols in public spaces.

8. Limitations

The authors noted several limitations and areas for improvement:

- The model showed signs of **potential overfitting** or data-specific learning challenges, evidenced by a **notable dip in validation accuracy** after the second epoch.
- The system currently relies on **binary classification** (“with mask” or “without mask”) and lacks the ability to distinguish between different **mask types**.
- The model may face difficulties in **challenging detection scenarios**, suggesting a need for more varied datasets and refined architectures to improve robustness against varied inputs.
- **Variability in performance** was observed through fluctuations in validation loss across subsequent training epochs.

3.3 Face Mask Detection System Using Deep Learning

1. Full Citation Details

- **Authors:** Madiha Zainul, Umarki Parveen, and **Ayantika Bera**.
- **Title:** **FACE MASK DETECTION SYSTEM USING DEEP LEARNING**.
- **Year:** **2021** (Project work duration 2021–2023).
- **Publication Venue:** A project work submitted for the degree of **Bachelor of Science**, Department of Computer Science, **Bangabasi Morning College, University of Calcutta**.

2. Specific Problem Identified or Addressed

The researchers addressed the difficulty of **identifying individuals not wearing masks** in real-time to ensure compliance with health policies during the rise of contagious diseases. They identified that **manual monitoring efforts are inefficient** and that traditional computer vision methods like Haar cascades and HOG have limited performance when dealing with variations in lighting, orientation, and occlusions.

3. Introduction/Background Summary

The context for this work is the intersection of computer vision, machine learning, and public health necessitated by global health crises. Automated systems are presented as a critical tool for providing **real-time alerts** for mask non-compliance. The background notes that while traditional handcrafted feature methods existed, the advent of **Convolutional Neural Networks (CNNs)** has made end-to-end mask classification significantly more accurate.

4. Research Motivation

The motivation was to assist in monitoring and enforcing safety measures to **reduce manual labor** and improve the efficiency of maintaining a safe environment. The authors aimed to contribute to public health by automating a repetitive task through intelligent systems.

5. Specific Objectives

- To develop a **real-time system** capable of identifying and localizing faces within video frames.
- To train a model that accurately classifies detected faces as **“Mask” or “No Mask”**.
- To provide immediate feedback through an intuitive **user interface** with visual overlays.
- To optimize the system for **real-time processing speeds** using hardware acceleration and efficient algorithms.

6. Methodology

- **Deep Learning Architecture/Model Used:** The system utilizes a pre-trained **MobileNetV2 model** for face detection and a **custom-trained mask detector model** for classification.
- **Dataset Details:** The research refers to utilizing labeled images of individuals with and without masks. It highlights datasets like the **Medical Masks Dataset** and **MaskedFace-Net** as robust sources for training.
- **Training Approach:** The model was implemented using **TensorFlow, Keras, OpenCV, and Imutils**. Preprocessing involved creating “blobs” from image frames, resizing them to **224x224 pixels**, and performing **batch predictions** with a batch size of 32.
- **Evaluation Metrics:** The system’s performance was evaluated based on **confidence probabilities** (threshold of 0.5) for each detection and overall **accuracy and robustness** in real-time streams.

7. Main Contributions to Knowledge

- Development of an automated monitoring system that provides **clear visual indicators** (green for masked, red for unmasked) and probability values for each prediction.
- The implementation of a unique **gathering alert system**, which triggers a warning message if the number of bounding boxes (people) in a single frame exceeds **40**, signifying a social distancing violation.
- Creation of a pipeline that processes moving objects and can generate a **CSV file** documenting the detected images.

8. Limitations

The authors noted several apparent limitations and areas for further development:

- The system’s robustness may be challenged by **challenging lighting conditions**, significant variations in **face orientation**, and **occlusions**.
- While performing well in tests, there is a noted need to improve accuracy specifically for **moving object detection** to ensure smooth video processing without lag.
- The project identifies the potential for future expansion to include more sophisticated **social distancing monitoring** and identification of specific violation types.

3.4 A Deep Learning-Based Tool for Detection of Face Mask and Measuring Body Temperature

1. Full Citation Details

- **Authors:** Yuxi Zhang, Ali Al-Ataby, and Fawzi Al-Naima.
- **Title:** A Deep Learning-Based Tool for Face Mask Detection and Body Temperature Measurement.
- **Year:** 2021.
- **Publication Venue:** Not explicitly listed in the provided excerpts, though the document is identified as a research paper focused on deep learning tools for pandemic safety.

2. Specific Problem Identified or Addressed

The researchers addressed the challenges of enforcing health protocols during the **COVID-19 pandemic**, specifically the need for mask compliance and normal body temperature in **overcrowded areas** like workplaces. They identified that **manual supervision** of these protocols results in **high labor costs** and increases the **risk of infection** for supervisory personnel due to close human contact.

3. Introduction/Background Summary

COVID-19 is a respiratory disease caused by the SARS-CoV-2 virus that has caused millions of deaths globally since its 2019 outbreak. Mask-wearing is considered one of the most effective ways to control transmission, and **high body temperature** is a primary symptom of infected patients. To mitigate spread, authorities require supervision at entrances to ensure compliance with mask-wearing and health norms.

4. Research Motivation

The motivation was to **save costs** associated with manual monitoring and **enhance safety** by reducing human contact between employees and supervisors. By automating the process, the authors sought to provide a safer, more efficient method for maintaining health protocols in company environments.

5. Specific Objectives

- To develop an **automatic, real-time tool** with a Graphical User Interface (GUI) for simultaneous mask detection and **non-contact temperature measurement**.
- To implement a system that provides **immediate feedback** at workplace entrances.
- To incorporate **Internet of Things (IoT)** functionality to send high-temperature alerts directly to administrators' smartphones.

6. Methodology

- **Deep Learning Architecture/Model Used:** The system uses a **two-stage detector** approach. It employs a prepared **SSD face detector** through OpenCV to extract the Region of Interest (RoI) and a **MobileNetV2** backbone for the actual mask classification.
- **Dataset Details:** The dataset includes two categories: “**with mask**” and “**without a mask**”. Images were sourced from online open-source platforms and generated via **Generative Adversarial Network (GAN)** algorithms. The authors also added “**confusing data**,” such as images of people covering their faces with their hands, to improve accuracy.
- **Training Approach:** The model utilized **fine-tuning**, where the pre-trained MobileNetV2 base layers were frozen and a **new custom head** with fully-connected layers was trained. Preprocessing included image augmentation and resizing. The system used the **Adam optimizer** (learning rate 0.0001) and **Binary Cross-Entropy** loss.
- **Evaluation Metrics:** The model was evaluated based on **training loss, training accuracy, validation loss, and validation accuracy** across multiple epochs.

7. Main Contributions to Knowledge

- The development of an **integrated, automated system** that handles both mask detection and temperature measurement in real-time.
- The implementation of a **brightness checking function** that analyzes grayscale pixel distribution to ensure detection accuracy in poor lighting conditions.
- Demonstration of a system that can **correctly identify hands** pretending to be masks, a scenario where many commercial systems fail.
- Achieving a high classification accuracy of approximately **98%** using an efficient, lightweight model.

8. Limitations

- The system struggles to distinguish between actual masks and **other objects** (such as books or mobile phones) used to cover the mouth and nose, which are sometimes incorrectly classified as masks.
- Current **image quality** and camera resolution are noted as non-ideal.
- The system is currently implemented as a **software prototype on a laptop**; the authors noted it requires dedicated hardware modules for market deployment.
- The authors suggested that at least **200 more pieces of confusing data** are needed to further refine the neural network’s ability to avoid false positives.

3.5 A Face Mask Detection System: An Approach to Fight with COVID-19 Scenario

1. Full Citation Details

- **Authors:** Ruchi Jayaswal and Manish Dixit.
- **Title:** A face mask detection system: An approach to fight with COVID-19 scenario.
- **Year:** 2022.
- **Publication Venue:** *Concurrency and Computation: Practice and Experience*.

2. Specific Problem Identified or Addressed

The researchers addressed the **scarcity of reliable face mask datasets** and the need for effective real-time algorithms to enforce safety protocols during the pandemic. They identified that existing research often focused on simple 2D or surgical masks, failing to account for more complex variations like **3D-printed, transparent, or printed masks**.

3. Introduction/Background Summary

The context is the COVID-19 pandemic, which originated in Wuhan, China, and spread to over 216 nations. Because the virus spreads through person-to-person contact, the WHO recommended **precautionary measures** such as wearing masks and social distancing. The authors posit that while deep learning has solved many computer vision tasks, creating a system that provides **high accuracy in real-time** remains a primary challenge for researchers.

4. Research Motivation

The work was motivated by the desire to assist society by providing **intelligent computer vision tools** that automate the detection of health norm violations. The authors sought to create a robust application that could operate effectively despite limited existing data, thereby contributing to **public health safety** and pandemic mitigation.

5. Specific Objectives

- To introduce a **unique real-time face mask dataset (FMDRT)** containing diverse mask types like 3D-printed and transparent masks.
- To propose a novel model, **CL-SSDXcept**, that combines advanced preprocessing, face detection, and feature extraction.
- To achieve the **highest possible accuracy** with minimum computation time and space.
- To compare the proposed model against standard models like MobileNetV2, VGG16, and InceptionV3 to validate its superiority.

6. Methodology

- **Deep Learning Architecture/Model Used:** The proposed model is **CL-SSDXcept**, which utilizes the **Caffe prototype SSD (Single Shot Detector)** for face localization and a fine-tuned **Xception model** for classification.
- **Dataset Details:** The authors introduced the **FMDRT dataset** (851 images: 551 with mask, 300 without). They also used the **RFMD dataset** (3,000 images selected) and a general **Face Mask dataset** (7,959 images) for comparative performance evaluation.
- **Training Approach:** Preprocessing involved **CLAHE (Contrast Limited Adaptive Histogram Equalization)** to enhance image quality. The model used the **Adam optimizer**, a **0.001 learning rate**, a batch size of 32, and was trained for **50 iterations/epochs** in a GPU environment.
- **Evaluation Metrics:** The system was evaluated using **precision, recall, accuracy, loss (error rate)**, and model generation time.

7. Main Contributions to Knowledge

- Creation of the **FMDRT dataset**, which is unique for including **3D-printed, transparent, skin-colored, and clown masks**.
- Development of the **CL-SSDXcept** technique, which achieved a high test accuracy of **98%** with a low loss of 0.05.
- Demonstration of a system capable of real-time performance at an average of **16 frames per second (fps)**.
- Evidence that the proposed model requires **less training time** than InceptionV3 while maintaining comparable accuracy.

8. Limitations

- The authors noted that **MobileNetV2**, while lightweight and fast, was ignored for this specific high-accuracy application due to its **poor accuracy** compared to the proposed model.
- Existing approaches are often **limited to 2D and surgical masks**, a limitation this study sought to overcome by including complex mask types.
- While the model performed excellently, the authors suggested that further research is needed to extend the system to **coughing and sneezing detection** using body gesture analysis.

3.6 Face Mask Detection Using Deep Learning

1. Full Citation Details

- **Authors:** Mr. Yogesh Sahu, Aayushi Korde, Anuj Malvi, Harshika Jawane, Prashu Vishwakarma, and Rashmi Maywad.
- **Title:** Face Mask Detection Using Deep Learning.
- **Year:** 2023.
- **Publication Venue:** *International Journal for Research Trends and Innovation (IJRTI)*.

2. Specific Problem Identified or Addressed

The authors addressed the need to protect public health from **respiratory illnesses**, specifically **COVID-19**, by ensuring people take necessary precautions. They identified that while mask-wearing is a widely adopted preventive measure, **enforcing mask policies** in high-traffic public spaces like airports, shopping malls, and healthcare facilities is a significant challenge that requires automated tools.

3. Introduction/Background Summary

With the outbreak of the **COVID-19 pandemic**, mask-wearing became an essential practice to halt the spread of the virus. Face mask detection has emerged as a critical computer vision and deep learning-based technology to support this practice. The background highlights that deep learning, particularly **Convolutional Neural Networks (CNNs)**, is uniquely suited for image classification and object detection tasks required to analyze faces in real-time video streams.

4. Research Motivation

The primary motivation was to provide a technological solution to help people **socialize and conduct business safely** during the pandemic. The researchers aimed to assist in enforcing safety measures by creating a system that identifies which masks are being worn and ensures they are being properly utilized to protect individuals from illness.

5. Specific Objectives

- To develop an **automated system** using computer vision and deep learning to identify mask-wearing status.
- To accurately detect if a mask is being **worn properly** (e.g., covering both the nose and mouth).
- To analyze real-time image or video streams and provide **intelligent alerts**.

6. Methodology

- **Deep Learning Architecture/Model Used:** The system utilizes a **Caffe-based face detector** for localization and a pre-trained **MobileNetV2** convolutional neural network for mask classification.
- **Dataset Details:** The dataset contains a total of **3,835 images**, divided approximately into half featuring people wearing masks and half not wearing masks.
- **Training Approach:** The model was developed using **Keras and TensorFlow**. Preprocessing included **resizing images** to a uniform size, converting them to grayscale, and normalizing pixel values. The MobileNetV2 network was **fine-tuned** by adding additional layers on top and training the model end-to-end.
- **Evaluation Metrics:** The system's performance was assessed using **accuracy, precision, recall, and F1-score**.

7. Main Contributions to Knowledge

- Demonstration of a **real-time detection system** capable of successfully identifying face mask violations.
- Identification of **MobileNetV2** as an effective backbone for object detection and classification in pandemic safety tools.
- The creation of a **camera-agnostic, staff-friendly**, and easy-to-implement framework for automated surveillance in public spaces.

8. Limitations

- Deep learning models require **significant data and computational resources** for effective development.
- The implementation requires a **strong background in machine learning** and data science, which may be a barrier for some environments.
- A critical limitation noted is that many training datasets are **dominated by light-skinned individuals**, which can lead to the same pitfalls and biases found in traditional facial recognition technologies.
- The authors noted that the model may **not be suitable for all applications** or environments depending on available resources.

3.7 Deep Learning Based Safe Social Distancing and Face Mask Detection

1. Full Citation Details

- **Author:** Shashi Yadav.
- **Title:** Deep Learning based Safe Social Distancing and Face Mask Detection in Public Areas for COVID-19 Safety Guidelines Adherence.
- **Year:** 2020.
- **Publication Venue:** *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, Volume 8, Issue VII.

2. Specific Problem Identified or Addressed

The researcher addressed the difficulty of ensuring adherence to **COVID-19 safety protocols**, specifically wearing face masks and maintaining safe social distancing in public places. The paper identifies that **manual inspection** is labor-intensive and hard to enforce consistently across large gatherings or public spaces like shopping malls.

3. Introduction/Background Summary

The COVID-19 pandemic, first identified in Wuhan, China, in December 2019, created a global health crisis with high mortality rates compared to influenza. Because the virus spreads through person-to-person transmission, including from **asymptomatic carriers**, the WHO recommended wearing masks and maintaining a social distance of at least **2 meters**. As lockdowns were eased, automated inspection systems became necessary to monitor public gatherings and reduce the risk of transmission without relying on massive human resources.

4. Research Motivation

The work was motivated by the need to **minimize physical surveillance work** for police in containment zones and public areas. By providing automated alerts and public alarms, the system aims to save time, reduce the manual labor required for inspection, and ultimately contribute to **lowering the spread of the virus**.

5. Specific Objectives

- To propose an efficient computer vision approach for the **real-time automated monitoring** of social distancing and face mask compliance.
- To implement the system on **Raspberry Pi 4** to enable edge computing and real-time activity monitoring.
- To develop a system that sends **alert signals** to state police headquarters and triggers public alarms upon detecting violations.
- To build a robust model using a mix of **deep learning algorithms and geometric techniques** for detection, tracking, and validation.

6. Methodology

- **Deep Learning Architecture/Model Used:** The system utilizes a combination of **MobileNetV2** as the core backbone and the **Single Shot Detector (SSD)** framework. It also employs **triangle similarity techniques** and **Euclidean distance** measurements for calculating social distancing.
- **Dataset Details:** The study used a custom face crop dataset consisting of **3,165 images**. Images were annotated for two categories: “**mask**” and “**no mask**”.
- **Training Approach:** The approach involved **transfer learning and fine-tuning** using pre-trained **ImageNet weights**. The base layers of the MobileNetV2 were frozen while a new **fully-connected (FC) head** was trained. The model was trained for **1,000 steps** using the **Adam optimizer**, binary cross-entropy loss, a batch size of 32, and an initial learning rate of 1e-4 over 20 epochs.
- **Evaluation Metrics:** The system’s performance was measured by **accuracy** (achieving between 85% and 95%), **precision** (91.7%), **recall** (0.91), and **frame rate** (28.07 FPS).

7. Main Contributions to Knowledge

- Development of an integrated, **lightweight model** suitable for deployment on edge devices like the **Raspberry Pi 4**, balancing resource limitations with recognition accuracy.
- Introduction of an automated alert system that captures the face image of unmasked individuals and notifies authorities at **State Police Headquarters**.
- The application of **projective geometry** (triangle similarity) to calculate the distance of a person from the camera using assumed height and focal length, enabling distance calculation without specialized depth sensors.

8. Limitations

- While the SSD framework can detect multiple objects, the author notes that this specific implementation was **limited to the detection of a single person** for the distance calculation components.
- The distance measurement relies on an **assumed constant height** ($H = 165$ cm) for all individuals, which may lead to inaccuracies for people of significantly different heights.
- Traditional face detection may not be suited for scenarios where faces are occluded by **non-standard masks** like scarves or handkerchiefs unless specifically trained.
- The system requires a **threshold number of people** (e.g., more than 20) to be identified as violating norms continuously before a critical alert is triggered.

3.8 A Deep Learning-Based Approach for Real-Time Face Mask Detection

1. Full Citation Details

- **Authors:** Wadii Boulila, Muhanad Afifi, Ayyub Alzahem, Ibrahim Alturki, Aseel Almoudi, and Maha Driss.
- **Title:** A Deep Learning-based Approach for Real-time Facemask Detection.
- **Year:** 2021.
- **Publication Venue:** ArXiv (Research conducted at the Robotics and Internet-of-Things Lab, Prince Sultan University).

2. Specific Problem Identified or Addressed

The researchers addressed the **global health crisis caused by the COVID-19 pandemic**, noting that manual real-time monitoring of face mask compliance for large groups is a difficult and labor-intensive task. They specifically identified that **manual monitoring poses a health risk**, as human monitors must come into contact with hundreds of people daily, potentially acting as infection points themselves. Furthermore, they sought to address the **lack of systems that could detect if a mask is worn inappropriately** rather than just identifying its presence.

3. Introduction/Background Summary

Face masks have been adopted by many governments as an essential tool to safeguard public spaces and reduce the spread of the virus. While many public and work places are considered “hotspots” for infection, not all individuals are compliant with health mandates. Recent advancements in **Deep Learning (DL)** have shown excellent results in interpreting massive volumes of data accurately and quickly, making it a viable solution for automated enforcement of face-mask policies in real-time video.

4. Research Motivation

The work was motivated by the need to **safeguard public areas** while **eliminating human contact** during the monitoring process to protect health officials. The authors aimed to provide authorities and commercial spaces with a tool that is **accurate, fast, and implementable with minimum resources** on existing surveillance infrastructure.

5. Specific Objectives

- To develop a deep learning model capable of **detecting and locating facemasks** and determining if they are **appropriately worn**.
- To implement a solution that is **optimized for edge computing** to ensure real-time performance.
- To conduct a **comparative analysis** of the proposed model against state-of-the-art architectures like ResNet50, DenseNet, and VGG16.

6. Methodology

- **Deep Learning Architecture/Model Used:** The system utilizes **MobileNetV2**. It was chosen because it is a lightweight architecture suited for edge devices that can efficiently trade off accuracy and latency using global hyperparameters.
- **Dataset Details:** The authors compiled a complete dataset of **21,407 images**. It combined public datasets (**Face Mask Lite, Real-World-Masked-Face-Dataset, MaskedFace-Net, and Kaggle Face Mask Detection**) with their own collected data. The composition included 10,099 “with mask” images and 11,308 “without mask” images, including faces from different angles and masks worn incorrectly.
- **Training Approach:** The process involved **off-line processing** where images were converted to arrays and preprocessed using TensorFlow’s MobileNet input tools. **Data augmentation** techniques were applied, including geometric transformations, flipping, color modification, rotation, and width-height compensation. The model was trained using **Collaboratory Pro** with 32 batch sizes and **100 epochs**, taking approximately **12 hours** to complete.
- **Evaluation Metrics:** Performance was evaluated using **accuracy, precision, recall, and F1-score**, supported by a **confusion matrix** and classification reports.

7. Main Contributions to Knowledge

- Demonstration of a system achieving **99% accuracy, precision, and recall** in detecting mask-wearing violations in real-time.
- The development of a model that distinguishes between **masks worn correctly versus inappropriately**, which is a novelty compared to many existing works.
- Empirical evidence that **MobileNetV2 is superior to ResNet50, DenseNet, and VGG16** for this task, particularly in **training time** (1 hour 26 minutes compared to over 18 hours for VGG) and validation accuracy.

8. Limitations

- While the system is designed for real-world surveillance, the current results are based on **experimental laboratory datasets**; the authors noted that implementation in “real-world surveillance cameras in public areas” is a future objective.
- The system uses an **alert system (winsound.Beep)** for non-compliance, which may require further refinement for large-scale, high-noise environments.
- The model requires a **lengthy training time (12 hours)** even for a lightweight architecture, which might impact rapid iterative updates if the dataset expands significantly.

4. REAL-TIME DEPLOYMENT AND APPLICATION STUDIES

4.1 Evaluating the Real-World Application Efficacy of MobileNet Models

1. Full Citation Details

- **Authors:** Sara Bouraya and Abdessamad Belangour.
- **Title:** **Evaluating the Real-World Application Efficacy of MobileNet Models.**
- **Year:** **2024** (Published September 28, 2024).
- **Publication Venue:** *International Journal of Engineering Trends and Technology (IJETT)*, Volume 72 Issue 9.

2. Specific Problem Identified or Addressed

The researchers addressed the challenges surrounding the **practicality and trustworthiness of object detection models in real-life settings**, where complex situations like varying lighting and weather conditions often degrade performance. Specifically, they sought to evaluate how lightweight models handle **dynamic environments**—such as urban or rural settings—while maintaining a balance between **computational efficiency and accuracy**.

3. Introduction/Background Summary

Object detection is a cornerstone of technologies like autonomous driving, security surveillance, and traffic management. The authors categorize detection models into **one-stage detectors** (e.g., YOLO, SSD), which prioritize speed, and **two-stage detectors** (e.g., R-CNN), which prioritize accuracy. Within this context, the **MobileNet family** is highlighted as a leader in efficiency for mobile and edge computing due to its use of **depthwise separable convolutions**. This paper builds upon the authors' previous research into object tracking, different “neck” models, and systematic video dataset classification.

4. Research Motivation

The motivation for this research was to **connect theory with practice** by moving beyond theoretical model structures to testing these models in **real-life situations**. The authors aimed to prove that lightweight convolutional networks could remain **resource-efficient yet effective** for real-time operations on devices with limited computational power.

5. Specific Objectives

- To explore the capabilities of **MobileNet and its variants** (V1, V2, and V3 Small/Large) in object classification and detection.
- To evaluate model performance under **diverse environmental conditions**, including different lighting and weather scenarios.

- To apply **transfer learning** and specific architectural alterations to increase model **adaptability** across various real-world settings.

6. Methodology

- **Deep Learning Architecture/Model Used:** The system uses a **Transfer Learning approach** with a MobileNet element as the core functional layer ("mobilenet_1.00_224"). The architecture includes an input layer for 224x224 RGB images, a **Global Average Pooling 2D layer** to reduce complexity, a **dropout layer (0.2 rate)** to prevent overfitting, and a final **Dense layer** for classification.
- **Dataset Details:** The models were trained on the '**Car Object Detection**' dataset, which consists of **1,176 handpicked and labeled images**. The data includes diverse scenarios: sunny days, low light (dawn/dusk), complete darkness (night), raindrops, mist, busy city streets, and deserted country roads.
- **Training Approach:** The dataset was split into **1,001 training images and 175 testing images**. The authors outlined specific alterations to the architecture and training methods designed to collapse feature maps into manageable forms while keeping vital information intact.
- **Evaluation Metrics:** The models were assessed using **training accuracy, validation accuracy, training loss, and validation loss** monitored over 10 epochs.

7. Main Contributions to Knowledge

- Verification that **MobileNetV3 Large** is the most robust variant, achieving a **97% validation accuracy rating** in diverse environmental tests.
- Demonstration that all four versions of MobileNet can handle **complex object detection tasks efficiently** on power-constrained devices.
- Evidence of **continuous development** in network design, showing how iterations from V1 to V3 have improved the ability of models to learn from training data and generalize to unseen environments.

8. Limitations

- **MobileNet (V1)** exhibited higher validation loss compared to **MobileNetV2**, suggesting it is slightly less optimized for generalization than its successors.
- The system setup is optimized for **highest computational efficiency** (membership probability), which the authors note involves a **tradeoff** where it may not handle dual probabilities simultaneously.
- Variations were observed in how different versions (specifically V1 vs. V2) **handle overfitting and generalize** to new data subset conditions.

4.2 Real-Time Monitoring of Personal Protective Equipment Adherence Using On-Device AI Models

1. Full Citation Details

- **Authors:** Yam Horesh, Renana Oz Rokach, Yotam Kolben, and Dean Nachman.
- **Title:** Real-Time Monitoring of Personal Protective Equipment Adherence Using On-Device Artificial Intelligence Models.
- **Year:** 2025.
- **Publication Venue:** *Sensors*.

2. Specific Problem Identified or Addressed

The researchers addressed the high risk of **hospital-acquired infections** caused by a lack of adherence to **Personal Protective Equipment (PPE)** protocols among medical staff. They identified that traditional monitoring methods—such as manual education or physical inspections—are **inefficient, sporadic, prone to human error**, and limited by the availability of human resources. Furthermore, existing AI solutions often focus solely on masks, require **constant network connections** to remote servers, and pose risks to **data privacy**.

3. Introduction/Background Summary

Medical staff are a primary vector for contagious infections within hospitals, and failure to follow precautions can extend patient stays by over **15 days**. While the COVID-19 pandemic accelerated the use of AI for mask detection, these technologies were not suited for the broader, **holistic monitoring** (masks, gloves, and gowns) required in clinical environments. This led to the need for a system that can operate **offline and in real-time** to provide immediate feedback without compromising security.

4. Research Motivation

The motivation was to **reduce the repetitive burden** of manual compliance monitoring and minimize the health risks to supervisory personnel who must otherwise come into contact with hundreds of people. The authors aimed to create a solution that balances **speed, resource utilization, and cost-effectiveness** while maintaining stringent data privacy standards for healthcare settings.

5. Specific Objectives

- To develop a **novel on-device, AI-based computer vision system** for real-time PPE monitoring.
- To implement a **holistic framework** that detects masks, gloves, and gowns simultaneously.
- To optimize the system for **edge computing** on a Raspberry Pi 5 to ensure rapid, offline processing.

- To validate the system's viability using **unseen medical staff** in a high-stakes environment like a **Cardiac Intensive Care Unit (CICU)**.

6. Methodology

- **Deep Learning Architecture/Model Used:** The system utilizes **Transfer Learning** with a lightweight **MobileNetV3** extraction model. It uses a three-step pipeline: **pose estimation (MoveNet)** to detect landmarks, **region cropping** (face, torso, palms), and **custom binary neural networks** for classification.
- **Dataset Details:** A custom dataset of **7,142 images** from 11 participants was used, covering various PPE combinations and ethnicities. After processing, this resulted in **5,674 face, 10,836 palm, and 7,029 torso images**.
- **Training Approach:** Models were trained for **10 epochs** using the **Adam optimizer** and binary cross-entropy loss via TensorFlow. The final model was optimized using **LiteRT** for deployment on a **Raspberry Pi 5**.
- **Evaluation Metrics:** The authors used **accuracy** (achieving 93–97% for individual items and 85.58% overall), **McNemar scores**, **Cohen's Kappa**, and **p-values** to assess statistical significance.

7. Main Contributions to Knowledge

- The creation of a **comprehensive monitoring framework** that accounts for gowns and gloves, moving beyond simple mask detection.
- Verification that an **on-device (edge) system** can achieve performance comparable to server-based solutions while providing **maximal user privacy** and protection against cyber threats.
- Demonstration of a **low-cost, scalable solution** that can be installed throughout medical facilities to prevent disease spread.

8. Limitations

- **Glove detection** was the weakest component, with its accuracy dropping during live deployment and its prediction significance becoming negligible ().
- The system misidentified **doctor's white coats** as surgical gowns and generated false positives for **slightly colored blue tops**.
- **Environmental confounders** such as height differences, sex, and background scenery (lighting and noise) significantly impacted accuracy.
- The model struggled with **irregular PPE usage**, such as one hand being gloved while the other was bare, or the use of **plastic bags** around gloved hands.

4.3 PPE-EYE: A Deep Learning Approach to Personal Protective Equipment Compliance Detection

1. Full Citation Details

- **Authors:** Atta Rahman, Mohammed Salih Ahmed, Khaled Naif AlBugami, Abdullah Yousef Alabbad, Abdullah Abdulaziz AlFantoukh, Yousef Hassan Alshaikhahmed, Ziyad Saleh Alzahrani, Mohammad Aftab Alam Khan, Mustafa Youldash, and Saeed Matar Alshahrani.
- **Title:** PPE-EYE: A Deep Learning Approach to Personal Protective Equipment Compliance Detection.
- **Year:** 2026 (Published January 11, 2026).
- **Publication Venue:** *Sensors* (MDPI).

2. Specific Problem Identified or Addressed

The researchers addressed the difficulty of ensuring **Personal Protective Equipment (PPE) compliance** on hazardous construction sites, where manual inspection is **time-consuming, error-prone**, and difficult to scale for large or complex projects. They specifically noted that in the Kingdom of Saudi Arabia, many construction sites are in **remote areas** where compliance is often ignored due to a lack of automated inspection systems.

3. Introduction/Background Summary

Construction sites are high-risk environments prone to injuries, falls, and hazardous material exposure, making equipment like helmets, vests, and safety glasses essential. Traditional monitoring relies on manual oversight, which is inconsistent. While AI-based monitoring has grown, most previous studies have not targeted the **specific environmental factors** (such as dust and sandstorms) or the massive scale of projects in the **Saudi Arabian construction industry**, such as the NEOM project and its future smart city, LINE.

4. Research Motivation

The motivation was to **reduce workplace accidents** and fatalities by providing construction managers with a tool for **continuous monitoring and instant feedback**. The authors aimed to fill a research gap by developing a system specifically suited for the requirements and environmental conditions of heavy construction in Saudi Arabia.

5. Specific Objectives

- To develop an automated, real-time system (PPE-EYE) using the latest deep learning for **PPE compliance monitoring**.
- To utilize the **YOLO11 architecture** to improve detection of small objects and performance in poor lighting or dusty conditions.

- To achieve a **higher mean average precision (mAP)** and **faster inference time** than existing models like Faster R-CNN or earlier YOLO versions.
- To create a **balanced and augmented dataset** to improve the model's robustness and accuracy.

6. Methodology

- **Deep Learning Architecture/Model Used:** The system is based on the **YOLO11 architecture** (specifically **YOLO11x**), which avoids fixed anchors and is highly effective for detecting small PPE components.
- **Dataset Details:** The model was trained on the **CHVG dataset** (1,699 images, 11,603 objects), which was highly imbalanced. To resolve this, they used the **Synthetic Minority Over-Sampling Technique (SMOTE)** and integrated data from **Roboflow**, resulting in a balanced dataset of over **28,000 instances**.
- **Training Approach:** An **80:20 split** was used for training and testing. The model was trained for **21 epochs** using the **NAdam optimizer** with a batch size of 16 and a learning rate of 0.00001. Extensive **data augmentation** was applied, including flipping, rotation, zooming, and adjustments to brightness and exposure.
- **Evaluation Metrics:** The system was evaluated using **mAP50** (achieving 96.9%), **precision (0.9089)**, **recall (0.9075)**, **F1-score (0.90)**, and **inference time**.

7. Main Contributions to Knowledge

- Implementation of the **YOLO11x model** for PPE detection, achieving a state-of-the-art **inference time of 7.3 ms**, which is significantly faster than previous approaches (170 ms) using the same dataset.
- Development of a **tailored solution** for the Saudi Arabian construction context, considering local environmental challenges like dust.
- Demonstrated the efficacy of combining **SMOTE with extensive data augmentation** to balance the CHVG dataset, leading to a 7% improvement in mAP50 over earlier versions like YOLOX.
- Creation of a **software prototype with a User Interface (UI)** that handles real-time CCTV feeds and generates incident alerts.

8. Limitations

- **Hardware Dependency:** Achieving low inference times requires **high-end edge devices** with powerful GPUs; performance may degrade on standard hardware.
- **Scalability:** The system has currently only been tested at a relatively **small scale** in an academic laboratory environment.
- **Limited Classes:** The current model only detects **main PPE** (helmets, vests, glasses) across eight classes; it does not yet account for other tools like gloves.

- **Environmental Variability:** Future work is needed to reevaluate the model under a wider range of conditions, including **night shifts** and diverse indoor/outdoor settings.

5. COMPARATIVE ANALYSIS

5.1 Comparison of Face Mask Detection Approaches

Table 1: Comparative Analysis of Face Mask Detection Systems

The following table provides a comparison of the eight face mask detection papers:

Author & Year	Deep Learning Architecture	Dataset Size & Source	Reported Performance	Main Strength	Primary Limitation
N. Anusha et al. (2023)	MobileNet, YOLOv5, and ResNet-50	3,853 images; Face Mask Lite, Real-World Masked Face, MaskedFace-Net	99% accuracy (MobileNet); 55% (YOLOv5)	Integrated system for both mask detection and social distance monitoring	Difficulty recognizing blurry images or faces covered by hands/scarves
Ratnam Dodda et al. (2025)	Custom CNN (Conv2D, MaxPooling2D, Dense layers)	Standard dataset (size not specified in excerpt)	>90% validation accuracy; nearly 100% training accuracy	Cloud-based deployment (Google Colab) removes need for local hardware	Fluctuations in validation metrics suggest potential overfitting
Ayantika Bera et al. (2021)	MobileNetV2 (face detection) + Custom Mask Detector	1,376 images (690 masked, 686 unmasked)	98.7% accuracy	Real-time alerts for social gatherings (if >40 bounding boxes detected)	Vulnerability to challenging lighting and face orientations
Yuxi Zhang et al. (2021)	MobileNetV2	Open-source platforms + GAN-generated	~98% accuracy	Holistic tool combining mask detection with body temperature	System incorrectly classifies non-mask objects (books/phones) as masks

Author & Year	Deep Learning Architecture	Dataset Size & Source	Reported Performance	Main Strength	Primary Limitation
Ruchi Jayaswal & Manish Dixit (2022)	CL-SSDXcept (SSD face detector + Xception classification)	images FMDRT (851 images); RFMD (3,000); Face Mask (7,959)	98% accuracy; 0.05 loss	Includes diverse mask types like 3D-printed, transparent, and clown masks	Standard MobileNetV2 found to have insufficient accuracy for their specific task
Yogesh Sahu et al. (2023)	Caffe-based detector + fine-tuned MobileNetV2	3,835 images (approx. 50/50 split)	Described as “high accuracy” (not explicitly numbered in results)	Staff-friendly, camera-agnostic, and easy to implement	Datasets are biased toward light-skinned individuals
Shashi Yadav (2020)	MobileNetV2 + SSD	3,165 images (custom face crop dataset)	85% to 95% accuracy; 28.07 FPS	Edge computing deployment on Raspberry Pi 4 with police HQ alerts	Social distancing accuracy relies on an assumed constant person height (165cm)*
Wadii Boulila et al. (2021)	MobileNetV2	21,407 images (Combined public + own dataset)	99% accuracy	Can determine if a mask is appropriately worn (vs. just present)	Extensive training time (12 hours) required

5.2 Comparison of Real-Time Application Studies

Table 2: Comparative Analysis of Real-Time Deployment Studies

The following table compares the real-time application and Personal Protective Equipment (PPE) papers.

Author & Year	Application Domain	Technology / Approach	Real-World Deployment Considerations	Key Contribution	Primary Limitations
Sara Bouraya & Abdessamad Belangour (2024)	General Object Detection (specifically vehicles)	Transfer learning using MobileNet variants (V1, V2, V3 Small/Large)	Designed for edge computing and dynamic settings requiring real-time operation with limited resources.	Demonstration of how iterations from V1 to V3 improve generalization in adverse environments (varying lighting/weather)	Highest computational efficiency setup handles membership probability but cannot manage dual probabilities simultaneously .
Yam Horesh et al. (2025)	Holistic PPE (masks, gloves, and gowns) for healthcare	MobileNetV3 extraction model combined with MoveNet pose estimation and custom binary classifiers.	Optimized for offline inference on a Raspberry Pi 5 to ensure data privacy and zero network dependency.	Implementation of a privacy-preserving computer vision system that monitors more than just masks to reduce nosocomial infections.	Significant accuracy reduction in glove detection during live use; misidentifies medical white coats as surgical gowns.
Atta Rahman et al. (2026)	Full PPE (helmets, vests, glasses) for construction	YOLO11x architecture using SMOTE for dataset balancing and NAdam optimization.	Targets remote construction sites; requires high-end edge devices with powerful GPUs to maintain low inference times.	PPE-EYE system achieves a state-of-the-art 7.3 ms inference time , significantly faster than previous Faster R-CNN approaches.	Class selection is limited to “main PPE” (no gloves); performance in night shifts and various weather extremes remains a future goal.

Deployment Considerations

- **Edge Capability:** Both Bouraya (2024) and Horesh (2025) emphasize the use of **MobileNetV3** for its lightweight properties on single-board computers like the Raspberry Pi.
- **Privacy vs. Speed:** Horesh (2025) prioritizes **offline processing** for clinical data security, while Rahman (2026) focuses on high-speed **YOLO11x** detection to handle the massive scale of Saudi Arabian infrastructure projects like NEOM.
- **Robustness:** Bouraya specifically identifies that real-world efficacy must account for **environmental challenges** like raindrops and low lighting. Horesh and Rahman both note that **variations in height and gender** or specialized clothing (like doctor's coats) can confound detection models.

6. SYNTHESIS AND RESEARCH GAPS

6.1 Overview of Approaches and Performance

The sources indicate that the **MobileNetV2** architecture is the most prevalent approach for face mask detection, often integrated with **Single Shot Detectors (SSD)** or **Caffe-based** frameworks to facilitate real-time performance on edge devices. While some researchers utilize more complex models like **ResNet-50** or versions of **You Only Look Once (YOLO)**, lightweight architectures are consistently preferred for balancing computational efficiency with high **accuracy**, which typically ranges from **85% to over 99%** across the various studies. A notable innovation includes the **CL-SSDXcept** model, which combines an SSD face detector with the **Xception** network for classification to outperform standard MobileNet variants in accuracy while remaining relatively efficient.

6.2 Common Datasets and Findings

Common benchmarks cited in the sources include the **Real-World Masked Face (RMFD)**, **Face Mask Lite**, and **MaskedFace-Net** datasets, which are frequently augmented or combined with **GAN-generated images** and custom collections to improve model robustness. These datasets vary significantly in size, from smaller, hand-labeled sets of approximately **1,300 images** to large compiled collections exceeding **21,000 images**. Despite high reported performance, the sources identify **recurring limitations**, such as decreased reliability in **challenging lighting conditions**, difficulty recognizing faces with **non-standard occlusions** (such as scarves or hands), and a tendency for models to incorrectly classify other objects—like **books or mobile phones**—as masks.

6.3 Identified Limitations and Research Gaps

Significant **research gaps** identified across these studies include a **lack of demographic diversity**, with some authors noting that training datasets are often dominated by **light-skinned individuals**, which can lead to biased outcomes in facial recognition components. Furthermore, while most models focus on the simple binary presence of a mask, fewer systems effectively addressed the **appropriate wearing** of masks (covering both nose and mouth) or the detection of **specialized mask types**, such as 3D-printed, transparent, or clown masks. Additionally, integrated systems for **body temperature measurement** or **social distance monitoring** often rely on simplified geometric assumptions—such as a **constant height** for all individuals—that may lead to inaccuracies when deployed in complex real-world environments.

- **Misclassification of Non-Mask Objects:** A recurring limitation is the system's inability to distinguish between actual protective gear and **household objects** (like books or mobile phones) used to cover the face. While some models can correctly identify a hand covering the mouth as "no mask," they still struggle with "confusing data" where any non-skin area is incorrectly categorized as a mask. A new project could focus on high-fidelity training to differentiate between medical-grade protection and **non-protective occlusions**.
- **Demographic Bias and Fairness:** Critics of current mask recognition technologies note that many training datasets are dominated by **light-skinned individuals**. This creates a gap in algorithmic fairness, as the systems may not perform with equal accuracy across diverse ethnicities or skin tones.
- **Environmental and Dynamic Robustness:** Performance consistently degrades under **challenging lighting conditions**, such as low light, complete darkness, or high-glare environments. Additionally, many systems struggle with **blurry images** or faces at high distances and varying orientations from the camera, which are common in real-world CCTV feeds.
- **Distinction of Mask Types and Proper Usage:** While most research focuses on a simple binary (mask vs. no mask), there is a need for models that can identify **non-standard face coverings** like scarves, mufflers, or handkerchiefs, which standard models often fail to recognize. Furthermore, while some models address "appropriately worn" masks, there is still room for improvement in detecting subtle violations, such as a mask **not covering the nose**.
- **Scaling and Hardware Integration:** Many current studies are limited to **laptop-based software prototypes**. There is a justification for research that optimizes these models for high-end **edge computing** (like YOLO11) to handle multiple camera feeds simultaneously with ultra-low latency while maintaining high precision on low-cost hardware.